

Research paper

Geostrategic study of the evolution of energy harvesting systems based on the patent activity of flywheels and regenerative shock absorbers

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ABSTRACT

This paper analyses the technological activity on energy harvesting and storage systems, such as flywheels and regenerative shock absorbers, for support of renewable energy and electric vehicle applications. The activity is analyzed based on the evolution of the patents over the last 20 years, based on databases such as Patbase and Derwent. The data reveal that less than 50 % of the total flywheels patents and utility models are granted. China is identified as the leading innovator in this area, followed by the United States, Japan, and Germany, which collectively account for 84 % of all patent families. A ranking of related companies and institutes is dominated by Chinese companies such as CANDELA (SHENZHEN) TECHNOLOGY INNOVATION CO., SHENYANG WEIKONG NEW ENERGY TECH CO., followed by German and American companies such as LUK LAMELLEN, VYCON, Siemens, and Caterpillar Inc. The leading regions on regenerative shock absorbers technologies are Germany, China, USA, and the European Patent Office (EPO). The study shows a leadership of Germany accounting for 39 % of the priority patents. The study also identified the top 20 applicants with the highest number of patent families, which included European companies such as SCHAEFFLER AG, HYDAC, VOLKSWAGEN AG, and Continental Automotive GmbH.

1. Introduction

Currently, energy harvesting and storage systems are gaining interest due to their high potential to improve the efficiency and life cycles of a wide variety of systems. The random nature of primary energy availability leads to fluctuations in energy surplus and shortage; its intermittent nature means that constant energy generation is not guaranteed. This has prompted research into storing energy such that it can be released as required (in the same form in which it was collected or in a different one,) in response to power fluctuations. Energy storage implementations and selections are made considering economic and technical factors and they present a critical role in ensuring the energy future. Fig. 1 shows the evolution of the energy storage installations predicted to 2030, in which there is a clear growing trend in all developed countries. Regionally, the Asia-Pacific (APAC) will lead storage construction in megawatts by 2030, but the Americas will lead by

megawatt hours because US storage plants generally have more storage hours. Europe, the Middle East and Africa (EMEA) is currently lagging behind due to a lack of specific storage incentives and policies, which is surprising considering Europe's ambitious climate targets.

Over the past few decades, advancements in technology have brought about a revolution in the transportation and energy generation industry. Flywheels and regenerative shock absorbers are two technologies that have gained attention in recent years due to their potential to improve mechanical energy storage efficiency. A flywheel is a mechanical device that spins at high velocities and stores kinetic energy as a rotational motion. An electric vehicle (EV) exemplifies the potential use of both technologies. A flywheel can be used to store the energy generated during braking or deceleration and take advantage of it for a more efficient acceleration, reducing the strain on the battery and increasing the vehicle's range. Additionally, flywheels can be used to provide a burst of power when needed, such as during overtaking or

List of abbreviations: FESS, Flywheel Energy Storage Systems; MB, Magnetic Bearings; AMB, Active Magnetic Bearings; PMSM, Permanent Magnet Synchronous Motor; RSA, Regenerative Shock Absorber; EV, Electric Vehicle; APAC, Asia and Pacific; EMEA, Europe Middle East and Africa; UPS, Uninterruptible Power Supply; JET, Joint European Torus; FGC, Flywheel Generator Converter; HMR, Hydraulic Motion Rectifier; USPTO, Patent Office of the United States; EPO, European Patent Office; WIPO, World Intellectual Property Organization; CIP, International Classification of Patents; CPC, Cooperative of Patents Classification.

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climbing hills, thereby enhancing the vehicle's performance. A regenerative shock absorber, on the other hand, is a type of damper that converts the kinetic energy generated during a vehicle's suspension travel into electrical energy that can be used to power other systems or can be stored in the vehicle's battery. By capturing and reusing this energy, regenerative shock absorbers can improve the vehicle's fuel efficiency and reduce emissions. The technological needs for flywheels and regenerative shock absorbers are driven by the desire to increase the efficiency of EVs and reduce their reliance on traditional fossil fuels. As the global demand for sustainable transportation options continues to grow, these technologies are becoming increasingly important for automakers and their suppliers. The benefits of flywheels and regenerative shock absorbers extend beyond just energy efficiency. They can also enhance the overall driving experience by providing smoother acceleration and better handling, as well as reducing the vehicle's weight and improving its performance. Additionally, these technologies can help to increase the lifespan of other components and reduce maintenance costs, such as batteries and electric motors.

1.1. Applications of flywheels

Flywheel energy storage systems (FESSs) are classified based on power capacity and discharge time. New FESSs have significantly reduced energy losses and manufacturing costs and are able to work for several hours in a row (Amber kinetics, 2017). They offer high disabilities and fast responses and are not very sensitive to temperature or humidity. This makes them suitable for renewable energy production and distribution applications. Below, the main FESS applications and the technologies applied in the system components are described.

Frequency regulation is an important area of research and development for FESSs. Balancing generation and load in real time can be challenging in a renewable energy-based power grid, as the load on the system can fluctuate frequently due to the activation and deactivation of numerous individual loads, and even small deviations in frequency can cause damage to equipment (Kirby, 2005). FESSs effectively emulate the inertial response of traditional rotating machinery found in conventional power plants. They can help smooth out short-term fluctuations in power supply and demand, enhancing the overall stability of the grid. They are particularly effective in providing fast frequency response, which is crucial in maintaining grid stability in systems with high renewable penetration. A FESS can operate as a secondary power plant

by supplying and subtracting power to maintain a constant frequency. FESSs are well suited for this purpose because they can respond almost instantly, have a high energy/mass ratio, and a significantly longer lifespan in comparison to Li-ion batteries. Companies such as Beacon Power (Beacon power, 2010) and Helix Power (Helix power inc, 2024) are developing FESSs for grid applications. Research is also being conducted on control systems for FESSs to improve their frequency regulation capabilities in microgrids (Mahdavi et al., 2021) and wind power generation systems (Yao et al., 2017). FESSs are additionally being researched for use in renewable energy systems. Because wind and solar power can be unpredictable, FESSs are useful to smooth the power generation fluctuations and increase the efficiency of solar farms and wind turbines. There are many reviews and studies available on this topic (Mousavi G et al., 2017; Sebastián and Peña Alzola, 2012) but some recent research includes the development of nonlinear controllers for wind turbine outputs (Gayathri et al., 2017) and energy storage systems for hybrid PV-diesel systems (Ramli et al., 2015). These studies seek to improve the integration of FESSs into renewable energy systems. Many commercial FESSs are designed for use in uninterruptible power supply (UPS) systems. UPS systems based on flywheels offer a high quality and stability of the output, long lifespan, and low maintenance needs. Flywheels can be used in hybrid energy storage systems to smooth DC voltage fluctuations (Lashway et al., 2016). Companies such as Active Power Inc. (Active power inc.2024) and Calnetix/Vycons (Calnetix/Vycon, 2024) have developed flywheels for UPS applications with capacities of 2.8 kWh and 675 kW and 1.7 kWh and 450 kW, respectively. FESSs have been utilized in the context of nuclear fusion research, specifically at the Joint European Torus (JET) facility (Rendell et al., 2015). Two flywheel generator converters (FGCs) were used to provide supplementary power to the coils of the magnet load, with each flywheel capable of delivering 2600 MJ.

FESSs have also been used in transportation applications, where they can replace traditional powertrain systems due to their high power-to-mass ratio. Examples of such applications include flywheels for recovering braking energy in hybrid vehicles (Read et al., 2015), energy buffers for utility vehicles in urban traffic (Abrahamsson et al., 2014) or full-electric powertrains with a four-wheel drive that improved overall performance and extended the lifespan of batteries (Farrokhzad Ershad et al., 2021). FESSs have also been used in advanced locomotive propulsion and regenerative systems (Calnetix/vycon, 2021; Thelen et al., 2003) and heavy-haul transportation (Spiryagin et al., 2015, 2018),

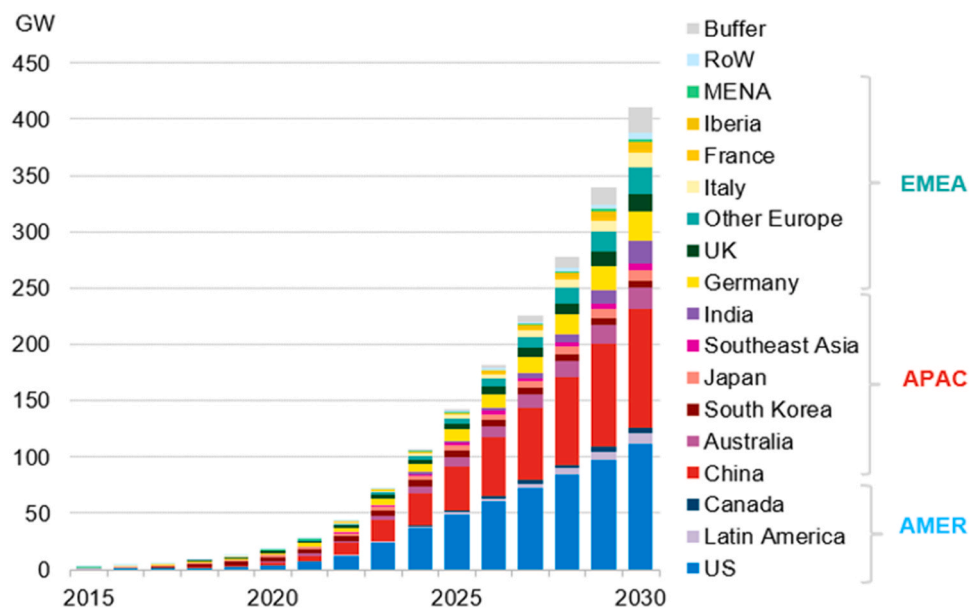


Fig. 1. Estimation of the evolution of global cumulative energy storage installations (BlombergNEF, 2024).

where they were found to be a favorable solution due to their specific power, cost efficiency, and service lifetime compared to batteries. In the marine transportation field, FESSs have been used as auxiliary components to improve the power quality of electrified ships. These systems have included high-efficiency motor/generators and magnetic bearings to reduce losses (Huynh et al., 2006), as well as diesel-mechanical tugboats (Kumar et al., 2018). A combination of flywheels and batteries has also been proposed to isolate load fluctuations on ships (Hou et al., 2018).

1.2. Components of flywheels

Most of the literature on FESSs focuses on the design of the different components and the configuration of the facilities in which FESSs are integrated. A complete review of the different technologies applied to FESSs and their applications was presented by Li and Palazzolo (Li and Palazzolo, 2022). Fig. 2 summarizes the technologies applied to the components of FESSs. The rotor is a critical component in a FESS because it accumulates the kinetic energy, and its design is therefore crucial for determining the energy capacity of the system. In most designs, the rotor is mounted on a shaft that also serves as the rotating component of the machine. The shaft is usually supported by bearings at both ends, which allow it to rotate smoothly and minimize friction. These bearings are typically designed to handle both radial and axial loads, ensuring that the rotor stays centered. The orientation of the rotor-shaft assembly can be either vertical or horizontal, and the materials commonly used to construct the rotor are composites or metals. Research on composite rotor design aims to improve the specific energy. A favorable relation between the strength and density of the rotor material is crucial. This requirement commonly leads to the use of composite materials, which have a high yield strength and low density; however, their weak radial strength is a limitation. Design optimization is often used, taking into consideration material combinations, shrink-fit allowances and thickness (Mittelstedt et al., 2018; Ramya et al., 2021; Rastegarzadeh et al., 2020; Uyar et al., 2020). Another solution is the use of several rims arranged in several layers that reduce the radial stress (Li and Palazzolo, 2022). High specific energy can be achieved by composite materials (50–100 Wh/kg), (Li et al., 2019) but the overall specific energy can be lower when using a metallic shaft. Steel flywheels have traditionally been seen as outdated technology with high-loss mechanical bearings. However, recent research has focused on increasing the specific energy and energy density through geometric optimization or novel configurations (Kale and Secanell, 2018). Due to their cost-effectiveness, recyclability and fabrication simplicity, interest in steel flywheels is re-emerging.

Another key part of the FESSs is the operational bearings. These can be either magnetic bearings (MB), which support the rotor, reducing friction loss and power loss, or fluid-film bearings, which require an additional lubrication system and are not suitable for vacuumed FESS.

These use both active and passive magnetic bearings. Active magnetic bearings (AMBs) can be homopolar or heteropolar and require the use of power electronics and a feedback regulator. Heteropolar AMBs are more common due to cost efficiency, and control is an important aspect in AMBs (Li et al., 2018; Peng et al., 2018). Passive magnetic bearings, on the other hand, require a higher complexity of design and assembly even though they do not need the feedback controller (Filion et al., 2013). A superior performance is found when using superconducting magnetic bearings (Cansiz, 2018; Karrari et al., 2018; Miyazaki et al., 2016; Morehouse et al., 2019) but their designs tend to be more complex and require cryogenic systems. Other types of bearings are hybrid systems that involve using passive and active magnetic bearings together (Xu et al., 2018). In some cases, passive magnetic bearings are used to provide stability in the axial direction, while more energy efficient and simpler radial repulsive magnetic bearings are proposed (Basaran and Sivrioglu, 2019; Han et al., 2013). A different type of hybrid system is the combination of magnetic and mechanical bearings (Sutherland et al., 2019).

The energy converter is a component of the system that operates as a motor or generator and is thus indispensable in FESSs. An electric machine, such as a permanent magnet synchronous motor (PMSM), converts kinetic energy into electricity and vice versa. The design of the electric machine is also relevant in adjusting the power rating of the flywheel. PMSMs are commonly used in FESSs due to their compact size and high efficiency. Factors to consider in the design include the size, grade, and number of poles of the magnets. One major design consideration is that the device must work in a vacuum environment, in which the friction is virtually zero and radiation is the only heat loss mechanism. Nagorny et al. (2005) provide a complete analysis on design solutions for PMSM in FESSs. Others, such as Schneider et al. (2017), proposed models for loss prediction and parametric optimization. Additional options for the energy converter are induction machines (Mir and Senroy, 2019; Yang et al., 2021), switch reluctance machines (Rocca et al., 2020; Sun et al., 2019; Zhu, Wang, and Cheng, 2021) or a bearingless machine (Sun et al., 2018; Yang et al., 2021; Yuan et al., 2016).

1.3. Applications of RSA

RSAs have mainly been applied in the automotive field as a part of the suspension system since they have potential use in energy savings and CO₂ emissions reduction. Other application fields are renewable energy production and active roads.

In the automotive industry, regenerative suspensions have emerged as a promising direction in vehicle research over the last two decades. These suspensions offer the potential to improve the dynamic behavior of the vehicle and restore the energy from road fluctuations into electricity. Several works have estimated the potential energy harvesting capabilities of vehicle suspensions and their impact on fuel efficiency. Zuo and Zhang (Zuo and Zhang, 2013) found that average tourism

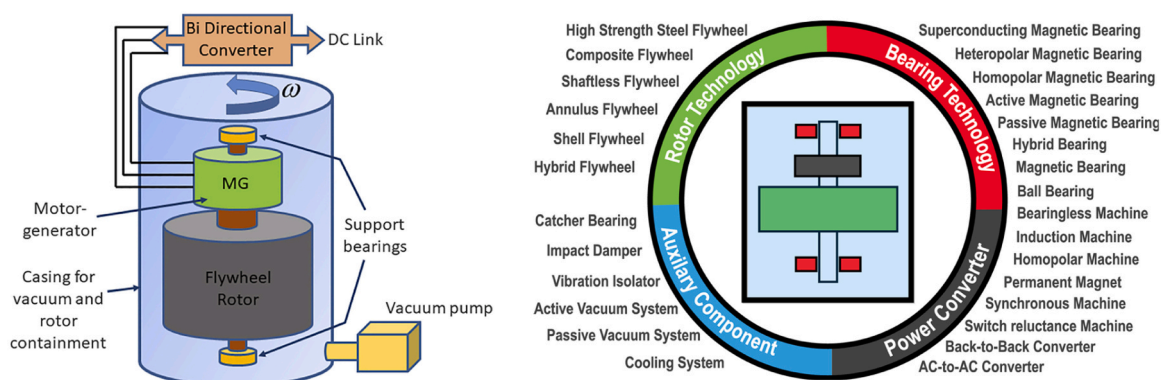


Fig. 2. General scheme of a FESS (left) and different technologies of the system components (right). Adapted from (Li and Palazzolo, 2022).

traveling at 97 km/h on well-maintained roads could generate between 100 and 400 W of potential energy. Audi (Audi's latest, 2023) reported that energy harvesting suspensions could reduce CO₂ emissions by 3 g/km for hybrid passenger cars on German roads and improve fuel economy by 0.7 L per 100 km. With an energy conversion efficiency of 75 %, this translates to a 3 % fuel efficiency improvement, according to BMW's data (Fairbanks, 2011). Levant Power (Levant power, 2023) reported up to 1 kW of potential fuel savings by partially recovering power loss in heavy-duty trucks and military vehicles. The power capacity of energy harvesting is dependent on the intensity of the vibrations. This indicates that more aggressive vibrations generate higher power and result in greater fuel savings. This effect is particularly noticeable in off-road vehicles and heavy-duty trucks. Harvesting systems based on vibration absorption have also been used for railroad systems (Tianchen et al., 2014). Although rail transportation does not produce high vibrations to be a potential source of significant amounts of energy, the energy harvested from these vehicles is used to feed safety systems and sensors, monitoring devices and signal lights, especially in remote regions where electricity supply is often scarce (Nagode et al., 2010, 2011). Therefore, a broad range of emerging applications exist in which harvested energy feeds low wattage systems (Ali et al., 2023). The use of this RSA can release the batteries from the continuous recharging of low wattage devices and increase the battery life cycle. Road pavements can also be a potential scenario for the use of RSA (Correia and Ferreira, 2021) based on installation in different roads and airport pavements. Such systems can combine vibration absorbers with different photovoltaic technologies applied to pavements. RSA systems are also applicable in renewable energy production devices such as wind turbines (Ong et al., 2020), in which the vibration reduction system is

crucial. Prudell et al. (2010) presented a design of a linear generator that uses the hydrodynamic kinetic energy of tides to generate electricity. The mechanism aims to generate 1 kW of electrical energy under average wave conditions off the coast of Newport, Oregon, USA, where waves reach approximately 1.5 m in height.

1.4. RSA mechanisms

An RSA system can convert the vehicle's linear oscillations on uneven roads into electricity through an electromagnetic circuit. Electromagnetic-based harvesters are popular in automotive energy harvesting suspensions due to their energy harvesting capability, high energy restoration efficiency, fast response and design simplicity. The RSA mechanisms are classified according to the method by which perpendicular fluctuations are transformed into electrical energy, and two types of electromagnetic harvesters are linear and rotary (Fig. 3). Several researchers have investigated both types of harvesters. Relevant mechanisms, configurations and working principles are presented in this section.

The first type of harvester is electromagnetic linear RSA, which uses electromagnetic induction to convert vertical vibrations into electricity. Unlike rotary electromagnetic RSA, which requires a transmission system to generate unidirectional rotation, the linear system generates electricity directly from vertical motion, eliminating the need for a transmission mechanism. Therefore, the electromagnetic harvester based on linear motion offers a significantly better behavior for power regeneration and can help save transmission losses. Zuo et al. (Zuo et al., 2010) created a prototype of a linear generator that utilized a harvesting damper to convert the energy of the fluctuations of the vehicle wheels

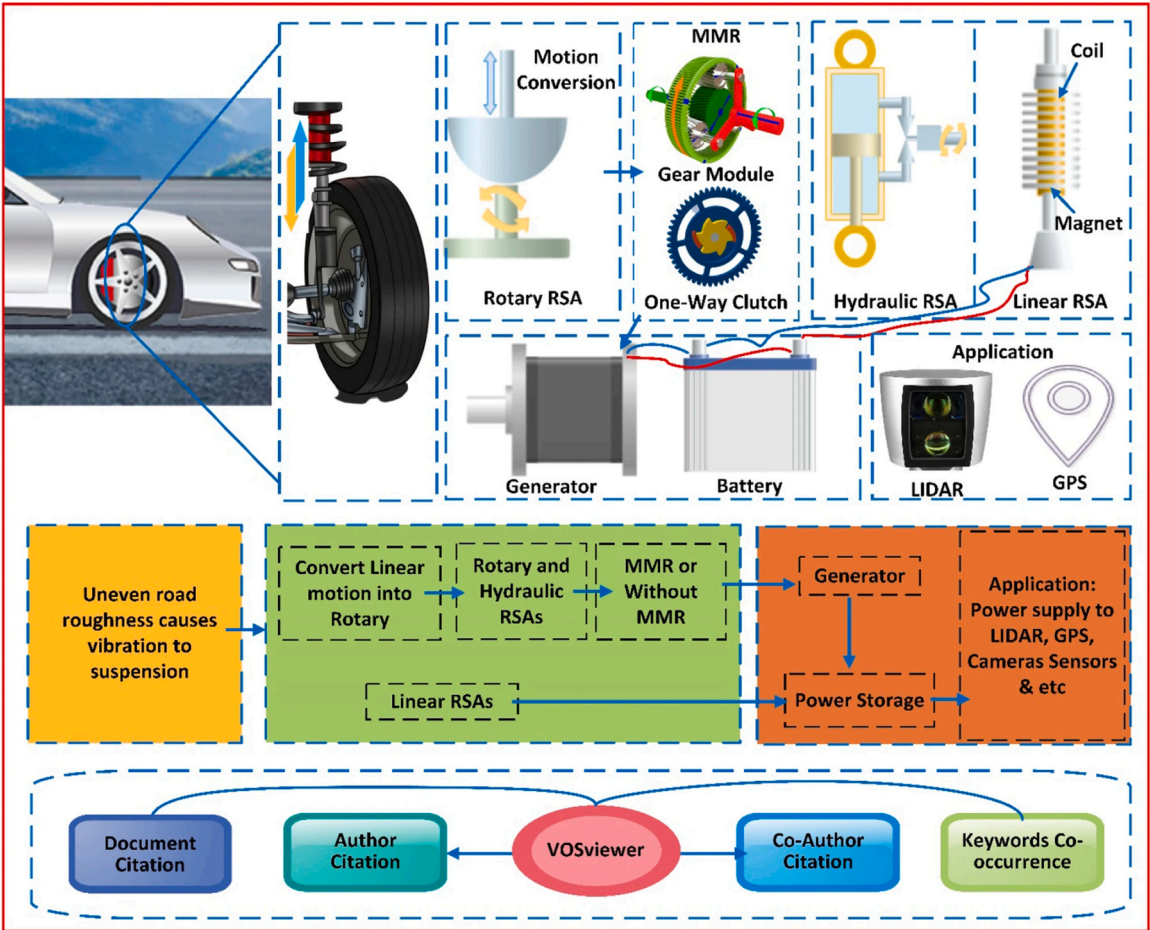


Fig. 3. Scheme of the main RSA technologies applied to suspension systems (Ali et al., 2023).

into electric energy. The linear-based harvester comprises the coil and magnet assemblies. The magnet configuration is made up of permanent magnets and high magnetically permeable spacers with a ring geometry, which are placed on a high reluctance rod. Similarly, Sapiński and Krupa (Sapiński and Krupa, 2013) proposed two linear generator designs that work as vibration energy harvesters in a linear motion rectifier damper. The first arrangement included two magnet assemblies, three spacers, and one coil section. The second one included three magnet assemblies, a coil with two sections and four spacers. Otherwise, self-powered actuation using energy harvesting from ambient vibrations has become a popular research area (Askari et al., 2017; Xinchun, Yonghu, Yi, Hui, and Jinping, 2015). Tang et al. (2014) proposed several designs to improve this type of system. In one these designs, radial magnetization magnets were introduced instead of metal separators. This increased the magnetic flux density passing through the coils, and therefore, the value of the generated volumetric power is higher. Electromagnetic suspension systems have shown promising results in improving the comfort, stability and manageability of vehicles with low power consumption ranging from 150 to 300 W depending on road conditions (Gysen et al., 2011). Electromagnetic generators are known for their high energy density and are easy to fabricate, in contrast to electrostatic and piezoelectric generators, which need an external voltage source or smart materials (De Araujo and Nicoletti, 2015).

Electromagnetic rotary harvesters convert the vibration energy of a car's perpendicular suspension vibration into electricity through a transmission system that translates perpendicular vehicle oscillation into rotary movement. Both mechanical and hydraulic transmissions are two prevalent methods for converting linear motion to rotary motion, which require different mechanisms and structures. According to various studies, mechanical-based designs are widely used in energy harvesting structures due to their straightforward construction and high efficiency. Various strategies based on mechanical concepts have been prototyped, devised, and thoroughly studied for regenerative suspension systems. These include rack-pinion mechanisms, ball-screw mechanisms, pulley-cable assemblies, algebraic screws, and other mechanical designs. Graves et al. (2000) suggested that the transmission gear ratio in rotary electromagnetic RSAs has the capacity to mechanically amplify absorption and regeneration. Zhang et al. (Zhang et al., 2007) presented a prototype of an RSA that utilized a ball-screw mechanism to produce 12 W of harvested power per damper and showed good efficiency when operating with low-frequency excitation but poor ride behavior for high-frequency excitation. Another RSA with ball-screw mechanisms was patented by Song et al. (2006) and achieved reasonably high-power conversion. However, it was found to exhibit poor ride comfort at high-frequency bandwidth. In addition, an RSA based on ball-screw transmissions combined with multiple regulated generators was proposed by Xie et al. (2017) to allow for continuous adjustment of the damping coefficient based on road conditions. An average electrical power of 32 W was regained with a 3 Hz frequency and 20 mm amplitude displacement input. Sabzehgar et al. (2014) applied an algebraic screw system. This was a mechanical assembly that transformed translational motion into rotational motion without requiring transmission gears and only using a planetary gear to ensure unidirectional rotation. Li et al. (Li et al., 2012) performed real road field testing combined with laboratory analysis on a novel design of a rack-pinion retrofitted RSA, achieving an overall power conversion efficiency of approximately 56 % under 0.5 Hz vibration frequency and 30 mm amplitude excitation. Furthermore, Zhang et al. (Zhang et al., 2016) designed a compact energy harvesting device utilizing track vibrations, which relies on a rack-pinion motion converter with rectified motion through one-way bearings. This system achieved an impressive efficiency of approximately 55.5 % when subjected to an oscillation of 6-mm amplitude and 2-Hz frequency.

The RSA based on hydraulic transmissions is a promising suspension system for energy recovery, despite it being relatively expensive in comparison to other technologies. For the hydraulic-based RSA to

generate electricity, it relies on the hydraulic fluid to transmit alternative movement of the cylinder to the motor. This rotational motion then powers an electrical generation circuit. To maintain stability in the performance of the hydraulic regenerative-based damper, certain components, such as motion rectifiers, hydraulic check valves or gas accumulators, are necessary for the system. Fang et al. (2013) proposed a hydraulic electromagnetic RSA that included an internal accumulator coupled with a rectifier. The proposed system achieved a regeneration efficiency of 16 %. Zhang et al. (2015) utilized an optimization technique based on a genetic algorithm to identify the optimal power regeneration trends of a hydraulic pumping regenerative suspension equipped with a hydraulic motion rectifier (HMR). This system collected an average power of 33.4 W for a simulated oscillating motion of 50 mm amplitude and 1.67 Hz frequency, with a theoretical hydraulic efficiency ranging from 70 % to 73 %. Li et al. (Li et al., 2014) created a regenerative damper that uses hydraulics and developed an HMR with four sets of check valves to correct the hydraulic motor's rotational direction. Through their experiments, they found that the hydraulic RSA utilizing the HMR reached a harvesting power of 248.8 W with an average of 114.1 W, which means, approximately, a 39 % regeneration efficiency at the optimum point. These results were obtained from a harmonic excitation of 8 mm and 2 Hz. Researchers at MIT (Chandler, 2009) created a hydraulic RSA to collect wasted vibration energy in a military truck. The harvester produced up to 1 kW of power per shock absorber on a standard road, which was sufficient to substitute the load of the alternator in heavy and military vehicles and could even power additional instruments. Demetgul and Guneş (2017) suggested a promising approach to generate electricity from linear motion through a hybrid RSA consisting of both electromagnetic and hydraulic mechanisms, which may lead to a balance that takes advantage of both systems' capabilities. Satpute, Singh, and Sawant (2014) and Satpute, Satpute, and Jugulkar (2017) created a hydraulic electromagnetic RSA with a horizontal linear generator that could generate a maximum power of 75–227 W at velocities ranging from 0.1 to 0.4 m/s. The system had a conversion efficiency of approximately 20 %.

1.5. Integration of machine learning and internet of things in enhancing energy storage technologies

The rapid advancement of energy storage technologies, coupled with the integration of machine learning (ML) and Internet of Things (IoT) systems, has revolutionized demand-side management (DSM) in smart grids. For the future development of energy storage systems (ESS), it is important to consider the intersection of these technologies, emphasizing the resilience of DSM systems against adversarial attacks. The robust design of IoT architectures, enhanced by deep learning models, plays a crucial role in maintaining the stability and reliability of smart grids.

As ESS are pivotal in balancing supply and demand, enhancing grid stability, and integrating renewable energy sources, traditional ESS solutions have been augmented with advanced ML algorithms to predict energy consumption patterns, optimize storage operations, and improve overall efficiency. For instance, Nawaz et al. (2020) proposed an intelligent integrated approach for efficient DSM, utilizing advanced metering infrastructure and forecasting frameworks to enhance grid performance. Similarly, the work by Kaur et al. (2019) demonstrated the use of a fleet of electric vehicles in a smart grid environment, employing an opportunistic-SDN-based edge-cloud framework to manage demand-response effectively. In recent years, the integration of IoT in energy management also has gained significant traction. IoT devices facilitate real-time monitoring and control of energy systems, providing granular data essential for efficient DSM. The study by Hashmi et al. (2021) highlights the implementation of IoT and cloud computing-based energy management systems, which significantly improve DSM by enabling real-time data processing and decision-making. Additionally, Qiu et al. (2023) explored the secure deployment of edge intelligence in

5 G-enabled Industrial IoT (IIoT), using a proof of learning consensus protocol to ensure data integrity and system security.

However, the practical implementation of these technologies faces several constraints. Real-world deployment must contend with the variability and noise inherent in energy consumption data. Techniques such as the use of continuous wavelet transform (CWT) for feature extraction, as employed in this study, help mitigate the effects of noise and enhance the robustness of the deep learning models. Previous research, including that by [Saxena et al. \(2022\)](#), has addressed noise in data through advanced filtering and preprocessing techniques, ensuring that the ML models receive clean and reliable input. Parameter tuning is another critical aspect of deploying ML models in energy systems. Hyperparameter optimization ensures that the models perform well under various operational conditions and adversarial scenarios. For example, the deep CNN model proposed by [Elsisi et al. \(2024\)](#) was optimized for hyperparameters to enhance its resilience against adversarial attacks, achieving a high accuracy of 94.06 %. This approach is corroborated by the work of [Chandra et al. \(2021\)](#), who utilized a transactive energy market framework to achieve decentralized coordination of DSM, ensuring optimal performance through continuous parameter adjustments. Moreover, the integration of ML and IoT in smart grids must address cybersecurity concerns. Cyberattacks on energy systems can lead to significant disruptions. The developed IoT architecture in this study employs CWT-based deep CNN to detect and mitigate such attacks, ensuring the integrity of the DSM system. Previous studies, such as those by [Taik et al. \(2021\)](#), have also emphasized the importance of secure communication protocols and federated learning to protect against adversarial threats.

Another problem of electricity grids integrated with renewable energy sources and power electronic-based generation sources, is the virtual inertia that affects the system's stability ([Liang et al., 2024](#); [Sockeye et al., 2020](#); [Yang et al., 2024](#)). The issue of virtual inertia can also be mitigated through the integration of IoT and machine learning technologies. IoT devices enable real-time monitoring and data collection from various grid components, providing the necessary input for ML algorithms to perform predictive analytics and dynamic control. Machine learning models can optimize grid operations, emulate virtual inertia, and enhance fault detection, demand response, and energy storage management. This integrated approach ensures improved grid stability, reliability, and efficiency by allowing rapid adaptation to fluctuations in generation and load, thus addressing the challenges posed by reduced physical inertia and high renewable energy penetration.

In summary, the synergy between advanced ML algorithms, IoT systems, and energy storage technologies holds immense potential for enhancing DSM in smart grids. By addressing real-world constraints, such as data noise and cybersecurity, and optimizing model parameters, these technologies can significantly improve the resilience and efficiency of energy management systems. This paper builds on the foundations laid by previous research to propose a robust IoT architecture capable of supporting resilient DSM against adversarial attacks, contributing to the development of more secure and reliable smart grids.

The study on flywheels and RSA as energy storage systems can have significant scientific and technological value in several aspects compared to studies on other more commonly studied energy storage systems, such as batteries or hydrogen. Some of these aspects are durability and high charge and discharge capacity, making them suitable for applications that require fast responses and high efficiency; the ability to be integrated into renewable energy generation systems; sustainability, as flywheels and regenerative dampers do not rely on scarce or toxic materials like some batteries; or robustness and resilience, as these systems may be more robust and less susceptible to failure under extreme conditions compared to batteries, making them attractive for critical applications.

With this background, this paper shows a study of the patent activity of flywheels and regenerative shock absorbers in the 2002–2022. For

both systems, the technology evolution, geostrategic positioning and technological leadership are evaluated based on the number of applications, grants and publications. In addition, the most influential companies and research centers are shown through the rankings of patent families and applications. This geostrategic study on the recent world patent activity offers a unique contribution by mapping the global innovation landscape and highlighting regional strengths and competitive advantages in these emerging technologies. Unlike typical scientific literature that focuses primarily on technical advancements and performance metrics, this study provides valuable insights into the geopolitical distribution of innovation, identifying leading countries and key players driving technological progress. This broader perspective is essential for policymakers, industry leaders, and researchers to make informed decisions regarding investment, research focus, and international partnerships.

2. Patent search

To retrieve patent information, more than 100 international patent databases were consulted covering the offices of the United States (USPTO), Europe (EPO), World Intellectual Property Organization (WIPO), and the national offices of France, Germany, United Kingdom, China, Korea, Japan, and Spain. WebScience and Scopus were also used to search for scientific articles related to the technologies studied. The review includes the patent documents collected on the mechanical energy storage systems of interest over the last 20 years. Specifically, the design of the search strategy combines both the use of key terms and the use of international patent classifications for each of the mechanical energy storage systems noted (Flywheels and RSA). With respect to the bibliometric or quantitative analysis on energy storage systems by Flywheels, from the set of 4971 patent and utility model families (11,370 publications) collected in the Patbase and Derwent databases. For its part, energy storage systems for shock absorbers have been collected 169 families of patents and utility models (599 publications). It is inferred that the area is in a development phase according to the trend of annual applications and grants.

The following are the various search criteria used for collecting the information using keywords combined with the codes of the International Classification of Patents (CIPs) and the Cooperative of Patents Classification (CPC) corresponding to the following groups and subgroups:

2.1. Area transport

B60 GENERAL VEHICLES

- o B60K6/08 Main engines comprising combustion engines and mechanical or fluid energy storage means [2006.01]
- o B60K6/10 Using a rechargeable mechanical accumulator, e.g., flywheel [2006.01]

B62 LAND VEHICLES (not moved by rails)

B64 AIRCRAFT; AVIATION; AERONAUTICS; SPACE FLIGHT

B65 TRANSPORT; PACKING; STORAGE; HANDLING THIN OR FILIFORM MATERIAL

2.2. Area physics

G01 METROLOGY; TESTING

G06 COMPUTING; COUNTING

2.3. Area electricity

H01 BASIC ELECTRIC ELEMENTS

H02 PRODUCTION, CONVERSION OR DISTRIBUTION OF ELECTRIC POWER

- o H02J1/00 Circuits for mains or distribution networks, DC [2006.01]
- o H02J1/00 Circuits for direct-current main or distributing networks [2006.01]
- o H02J1/14 Load balancing in a network [2006.01]
- o H02J1/16 Using dynamo-electric machines coupled to flywheels [2006.01]
- o H02J3/00 Circuits for mains or distribution networks, AC [2006.01] (H02J3/28 Load balancing arrangements in a network by means of energy storage [2006.01], H02J3/30 using dynamo-electric machines coupled to flywheels [2006.01])
- o H02J15/00 Electric energy storage systems (mechanical systems F01-F04; chemically H01M) [2006.01] (H02J15/007 involving storage as mechanical energy, e.g., flywheels)
- o H02K16/00 Machines with more than one rotor or stator [2006.01]
- o H02K7/02 Additional mass to increase inertia, e.g., flywheels [2006.01] (H02K7/025 for storing energy)

2.4. Mechanical area

F01 MACHINES OR ENGINES IN GENERAL; POWER PLANTS IN GENERAL; STEAM ENGINES

F02 INTERNAL COMBUSTION ENGINES; HOT-GAS OR COMBUSTION-PRODUCT POWER PLANTS

F03 FLUID MACHINES OR ENGINES; WIND, SPRING, OR WEIGHT MOTORS; MECHANICAL-POWER PRODUCING DEVICES OR MECHANISMS, NOT OTHERWISE PROVIDED FOR OR USING A SOURCE OF POWER NOT OTHERWISE PROVIDED FOR:

- o F03D9/10 Combination of wind motors with energy storage devices [2016.01]
- o F03D9/12 storing kinetic energy, e.g., using flywheels [2016.01]
- o F03G Spring, weight, inertia or analogous motors; devices or mechanisms producing mechanical power not foreseen elsewhere or using a source of energy not foreseen elsewhere (F03G3/00 Other motors, e.g., gravity or inertia motors [2006.01], F03G3/08 using flywheels [2006.01])

F15 ACTUATING DEVICES USING PRESSURE OF A FLUID; HYDRAULIC OR PNEUMATIC IN GENERAL

F16 TECHNOLOGICAL ELEMENTS OR COMPONENTS; GENERAL MEASURES FOR PRODUCING AND MAINTAINING EFFECTIVE FUNCTIONING OF MACHINES OR INSTALLATIONS; THERMAL INSULATION IN GENERAL

- o F16F15/30 Flywheels (F16F 15/16 has priority; suppression of vibrations in rotating systems using elastic elements or friction damping elements moving with the system F16F 15/12; general aspects of the rotating body F16C13/00, F16C15/00) [2006.01]
- o F16H33/00 Transmission systems based on the repeated accumulation and release of energy [2006.01]
- o F16H33/02 Rotary transmission systems with mechanical accumulators, intermittently connected flywheels [2006.01])

Y02E - REDUCTION OF GREENHOUSE GAS [GHG] EMISSIONS RELATED TO THE GENERATION, TRANSMISSION OR DISTRIBUTION OF ENERGY OR Y02E60/16 - Mechanical energy storage, e.g., flywheels or pressurized fluids

Y04 - INFORMATION OR COMMUNICATION TECHNOLOGIES HAVING AN IMPACT ON OTHER TECHNOLOGY AREASB60

3. Results and discussion

3.1. Flywheels

Below is a detailed analysis of the inventory of innovations related to flywheel-based mechanical energy storage systems over the past 20

years. A search returned a total of 4971 patent and utility model families, corresponding to 11,370 published documents. Of these, 53 % have been granted and are currently in effect, while the remaining 47.3 % are still pending approval. This signifies that less than half of the total number of patents and utility models have been granted. When looking at this breakdown by country, China leads the way in terms of innovation, followed by the United States, Japan, and Germany, which together account for 84 % of all patent families.

3.1.1. Technology evolution

The number of annual patent applications indicates the level of interest in a technology over time. It also allows for identifying the phase of the lifecycle of a specific technology area, which favors different types of innovation in each of its phases:

- The initial phases of a market coincide with the adoption and development lifecycle of a technology. It is often linked to more radical, application- or product-based innovation. The existing technology base is small, but the emergence of a "breakthrough" technology sets the stage for numerous subsequent developments.
- The middle phases of a market coincide with the mature phase. It is characterized by the constant emergence of incremental or process innovations that improve upon previous developments.
- Finally, during the declining and end-of-cycle phases, the emergence of new developments stagnates, and companies can only escape the "commoditization" of their market by reinventing themselves, innovating in their business model, or altering the structure of the sector, such as through a new "breakthrough" innovation that restarts the cycle.

Fig. 4 shows three variables for each year: the number of grants and applications published, which allows for evaluating the evolution of developments over the considered period of time, as well as the extent of publication in the markets of interest. There is a continuous increasing trend starting in 2006 in terms of annual innovation applications, with a maximum of 391 applications in 2019 (green bar), closely followed by 2021. It is important to note that the values for 2021 and 2022 may be higher than the reported values due to the hidden period of patent documents, which ranges from 12 to 18 months. From the perspective of publications (dark blue bar), there is also a growing trend until 2017, with a peak of activity at 657 publications, which then decreases to the present. Finally, annual grants show a continuous increasing trend over the past 20 years, with the maximum number of grants in 2020 at 462. These data are confirmed by the following figure showing the 13 national and international offices with the highest number of priorities per year, with China emerging as an absolute leader in knowledge generation with more than 90 % of innovations in recent years (as shown in Fig. 4 below). The United States, Japan, Germany, European Patent Office, and the United Kingdom are also seen to maintain a similar order of magnitude, but there is a new player with growing participation such as India. It is suggested that the area is in a mature but still developing phase based on the trend of annual requests and grants.

3.1.2. Geostrategic positioning

The geostrategic positionings allows for a double geographic analysis. On the one hand, the main countries and regions generating knowledge are identified, and on the other hand, the markets of interest are identified through the publication of these patents and utility models. Fig. 5 shows the 20 national and international offices ordered by the number of publications and indicates the total number of grants and requests in each office. China stands out as the absolute leader in knowledge generation with 53 % of priorities, followed in second to fourth place by the United States, Japan, and Germany with 14 %, 9 %, and 8 %, respectively. From a regional perspective, Asia-Oceania has generated 65 % of all innovations, while Europe and America represent 15 % and 14 % of total priorities, respectively. From the market

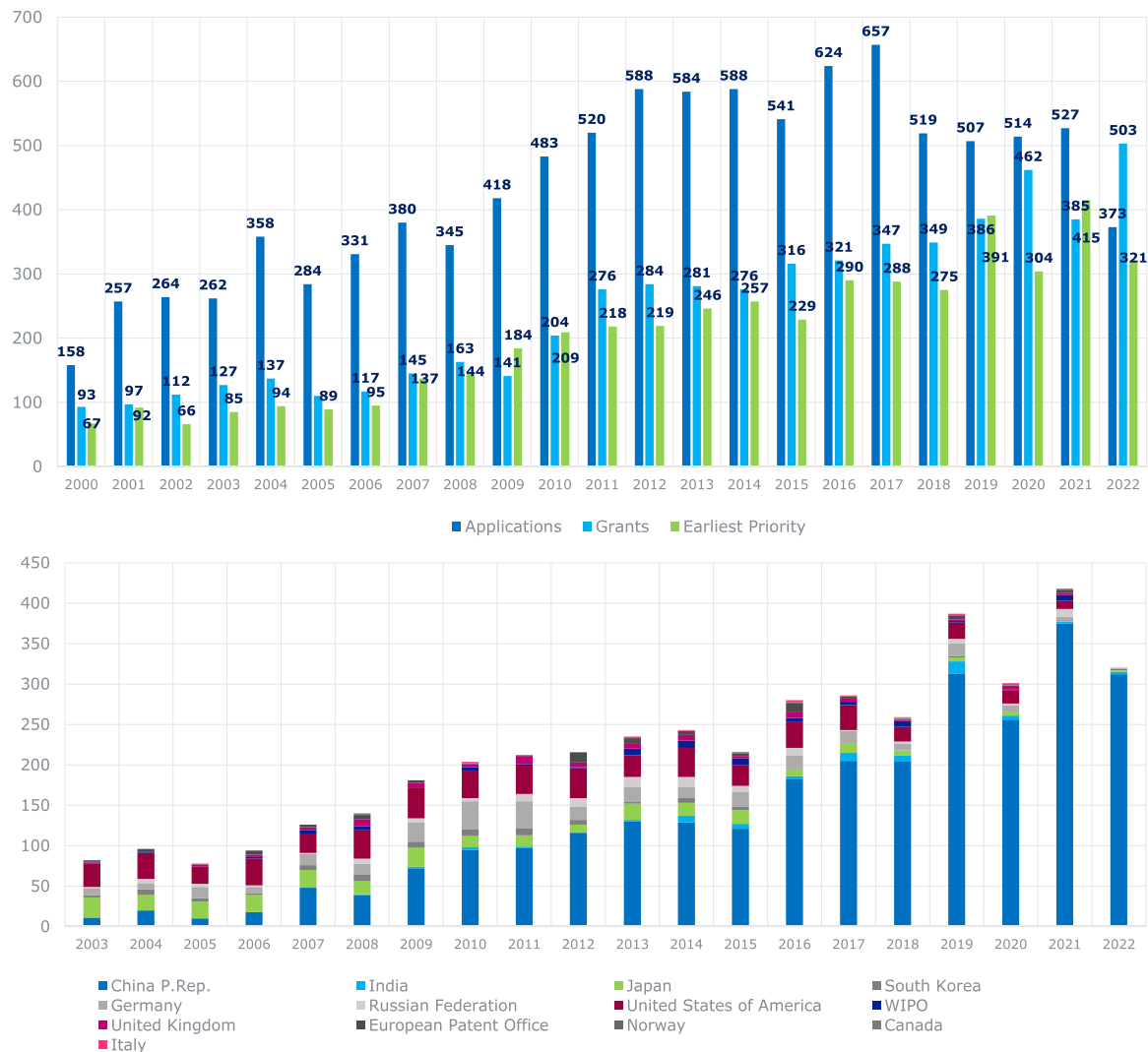


Fig. 4. Evolution of patent applications, applications and grants and details of the 15 national and international offices with the highest number of priorities.

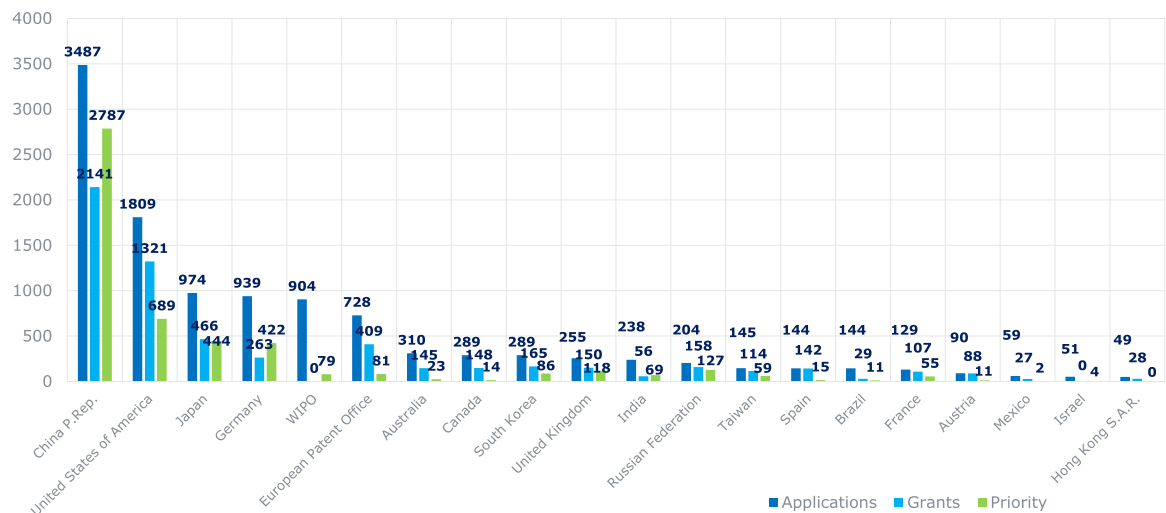


Fig. 5. Top 20 national and international offices ranked by publications, grants and patent applications.

perspective, the extent of the number of publications per office, China maintains its leadership with 29 % of total publications, followed by the United States, Japan, and Germany. From a regional perspective, Asia-

Oceania accounts for 45 % of the total, followed by Europe with 22 % and America with 20 %. On the other hand, the use of the World Intellectual Property Organization (WIPO) and EPO international offices

represents 14.1 %, which confirms the still very local character of the study area, focusing activity in the Asia-Oceania, Europe, and North America regions as the main markets of interest, as seen on the world map, with the most intense green color indicating the countries with the highest activity and showing minimal or nonexistent activity in Africa and Latin America (see Fig. 6).

Based on the validity of solutions, China represents 29 % of total publications and maintains its leadership in validity with 2005 documents in effect, followed by the United States with 889 documents in effect (50 %) out of a total of 1809 patent documents. On the other hand, if we consider the European Patent Office (EPO) and countries belonging to Europe, a total of 2499 publications are currently registered, of which only 40 % are in effect. Fig. 7 shows a balance of total publications in effect (green) or that have entered the public domain (red) in the 20 offices that currently have publications. Specifically, China and the United States comprise 51.5 % of the total number of documents in effect. Finally, a summary of the legal status of all collected documents shows that a total of 5385 documents (47.3 %) are currently in the application or granted phase and therefore in effect, while the remaining 52.7 % have already entered the public domain.

3.1.3. Technological leadership

Technological leadership shows the competitive environment and its relative position, allowing the identification of the most prolific applicants in the development of technologies related to the study area, as it is assumed that these regions are also the ones requesting the most patents. Below are two rankings of applicants: the first refers to the entities with the highest number of families, and the second ranking refers to entities with the highest number of publications within the collected sample. In both sets, entities stand out in different positions, as their position varies but new actors also appear in the rankings of families and publications, because some include a portfolio of innovations in the area of high magnitude but in most cases little extended as is the case of Chinese companies, universities, and research centers, versus a ranking by publications with entities with few families or innovations, but a very wide marketing strategy.

It is important to note that a technology may only be fully introduced in the market according to the participation of the applicants, where more than ¾ of the applicants are companies and the remainder correspond to universities and research centers and private inventors in nearly equal proportion. Fig. 8 shows the ranking of 20 entities by

magnitude of patent and utility model families, highlighting the participation of Chinese companies, universities and institutes in the top positions such as CANDELA (SHENZHEN) TECHNOLOGY INNOVATION CO., LTD (China), SHENYANG WEIKONG NEW ENERGY TECH CO. LTD (China), BEIJING HONG HUI INT ENERGY TECH DEVELOPMENT CO. LTD (China); DUNSHI MAGNETIC ENERGY TECH CO. LTD (China), INST ENG THERMOPHYSICS CAS although the German company LUK LAMELLEN stands out in second position with 45 innovations but widely spread with 376 publications. These companies are followed by the American company VYCON, which specializes in the development of energy storage systems with flywheels. In ninth position is SIEMENS (Germany), CATERPILLAR INC (US.), ROBERT BOSCH (Germany), KOREA ELECTRIC POWER CORP (KEPCO) (Republic of South Korea), GE (US), TOYOTA MOTOR CORP (Japan), HONDA MOTOR (Japan), HONEYWELL INTERNATIONAL (US).

Fig. 9 shows the second ranking of the 20 entities with the highest number of publications, with the German company LUK, already mentioned above, at the top, followed by companies from the health sector such as ETHICON ENDO SURGERY INC (US) and COVIDIEN AG (Ireland), where only one invention has been widely spread. In 4th position is AMBER KINETICS INC (US), another company specialized in energy storage services with Flywheel, followed by the aforementioned companies. In 8th position is a multinational company from the aeronautical sector such as BOEING CO (US), others from the energy, industrial automation, security and automotive sectors such as BEACON POWER LLC (US), ABB LTD (Switzerland), JFE HOLDINGS INC (Japan), ENRICHMENT TECHNOLOGY US, Inc. (US), RAYTHEON (Canada) and FORD MOTOR (US). The different American and European companies are widely spread; therefore, there are innovations that may represent a commercial barrier for the introduction of the solutions in the area subject of the invention in those markets.

The main fields of application according to the flywheel classification database are notable in that more than 51 % of the families fall under the field of electrical machinery, followed by transport with 22 %, machines, pumps, turbines with 23 % and mechanical elements with 20 %. Other fields show activity of less than 200 innovations (Fig. 10).

3.2. Regenerative shock absorbers

The following is a brief quantitative and qualitative analysis of the inventory of innovations relating to mechanical energy storage systems:

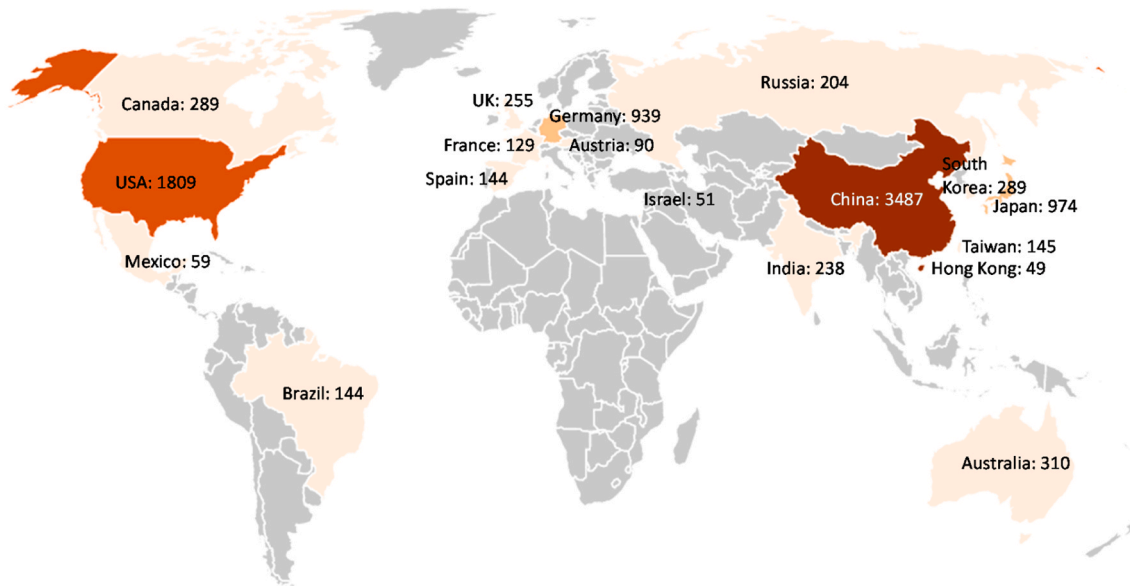


Fig. 6. World map of major national offices based on number of applications.

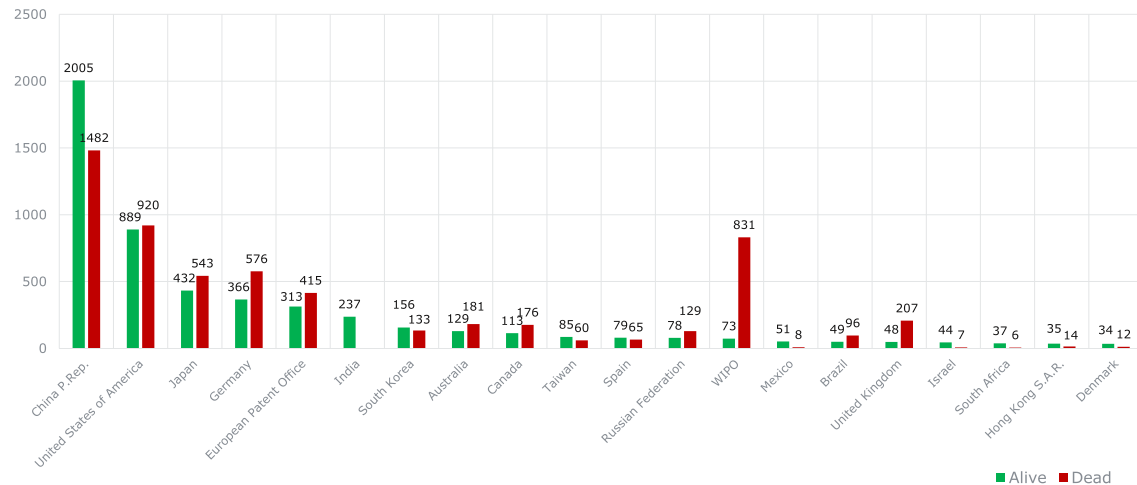


Fig. 7. Balance of the 20 national and international offices by validity of the patent documents and distribution of the total legal status of the documents collected.

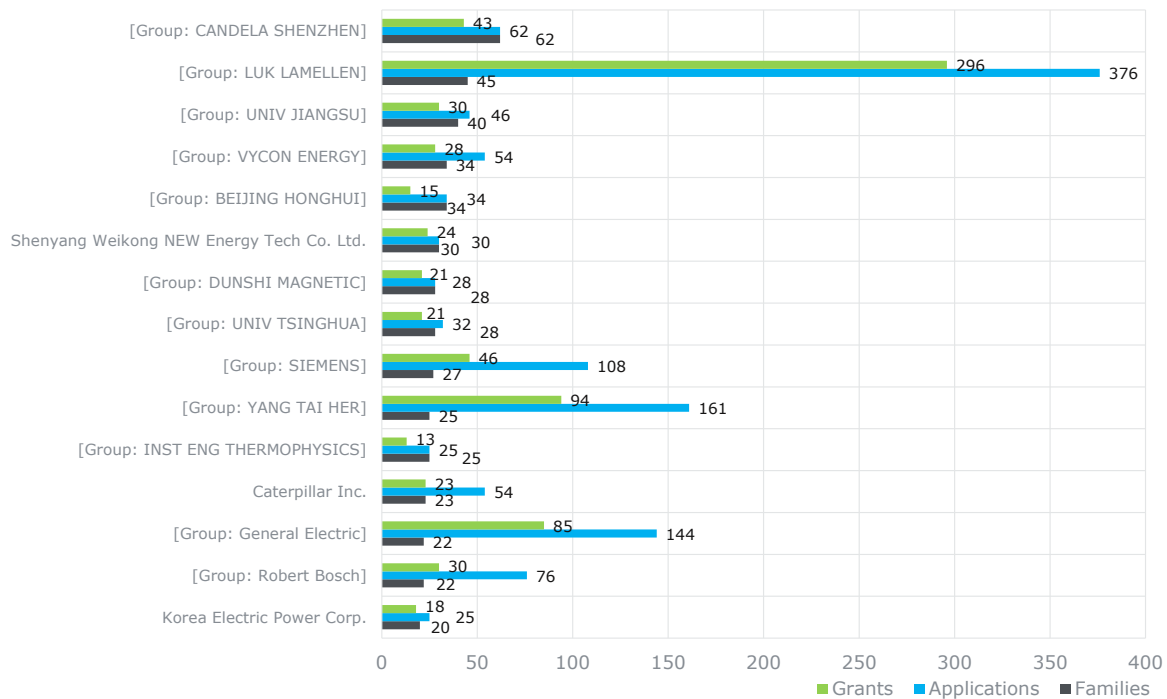


Fig. 8. Top 20 entities with the highest number of patent families.

shock absorbers (unlimited years). The review yields a total of 171 patent and utility model families, resulting in 616 publications, of which 55 % have been granted and 47.3 % are currently in effect. The dominance of Germany, China, the USA and the European Patent Office (EPO) in generating solutions in this area stands out.

3.2.1. Technology evolution

A variable trend is observed in terms of annual innovation applications over the last two decades, with a maximum of 15 applications in 2021 (green bar in Fig. 11 above). Based on publications (dark blue bar), a variable trend is also observed over the last 20 years, with the highest activity recorded with 45 publications in 2016. Finally, the annual concessions also denote a variable but increasing trend, reaching the highest number of concessions in the last 10 years; specifically, in 2020 and 2021, the highest number of concessions is recorded. It is important to note that the years 2021 and 2022 are data that should be taken with

reservations since the values may be higher due to the hidden period in the application of patent documents, which ranges between 12 and 18 months. Fig. 11 (below) shows the 10 national and international offices with the highest number of priorities per year, confirming that China and Germany have been the absolute leaders in the generation of knowledge in recent years. It is suggested that the area is in a development phase according to the trend of annual applications and concessions.

3.2.2. Geostrategic positioning

The top 20 offices for the generation and publication of inventions in the area are shown in Fig. 12, as well as the number of documents granted by each office. Germany is the leader in knowledge generation with 39 % of the priorities, followed in second and third position by China and the USA with 34 % and 11 %, respectively. From a regional perspective, Asia and Europe represent 36.1 % and 49.1 % of the total

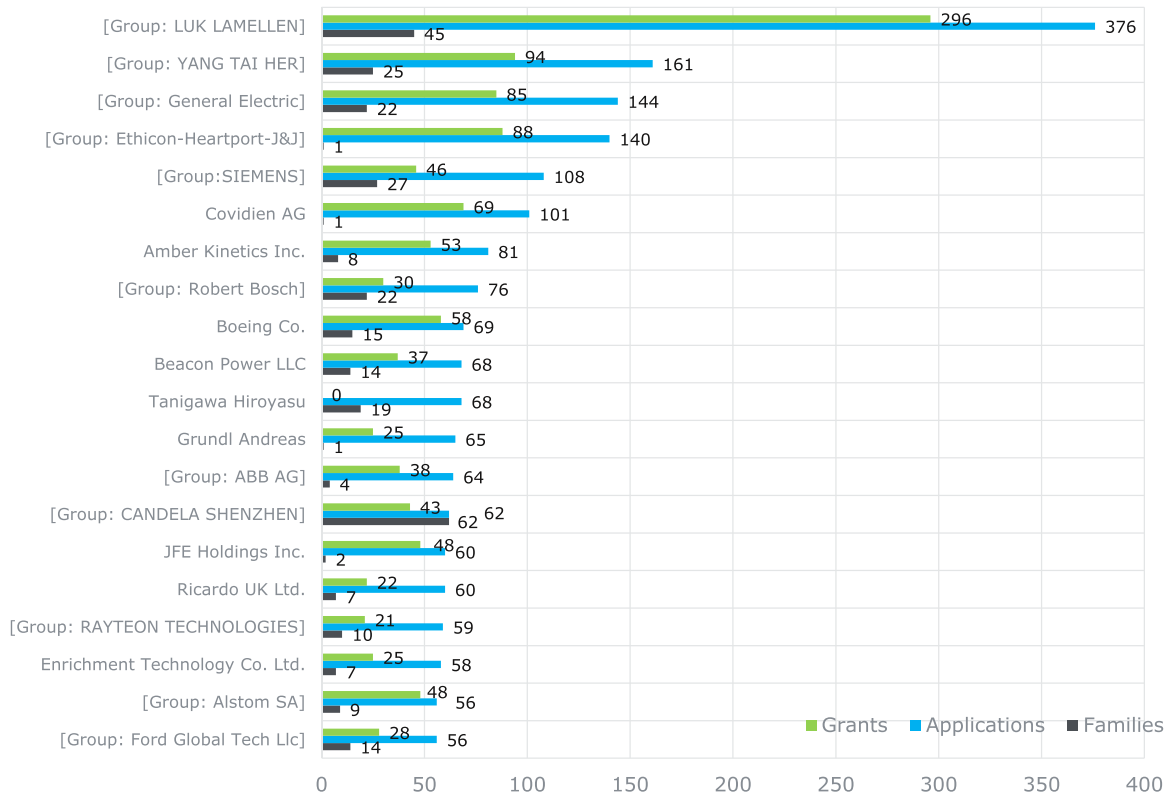


Fig. 9. Top 20 entities with the highest number of applications.

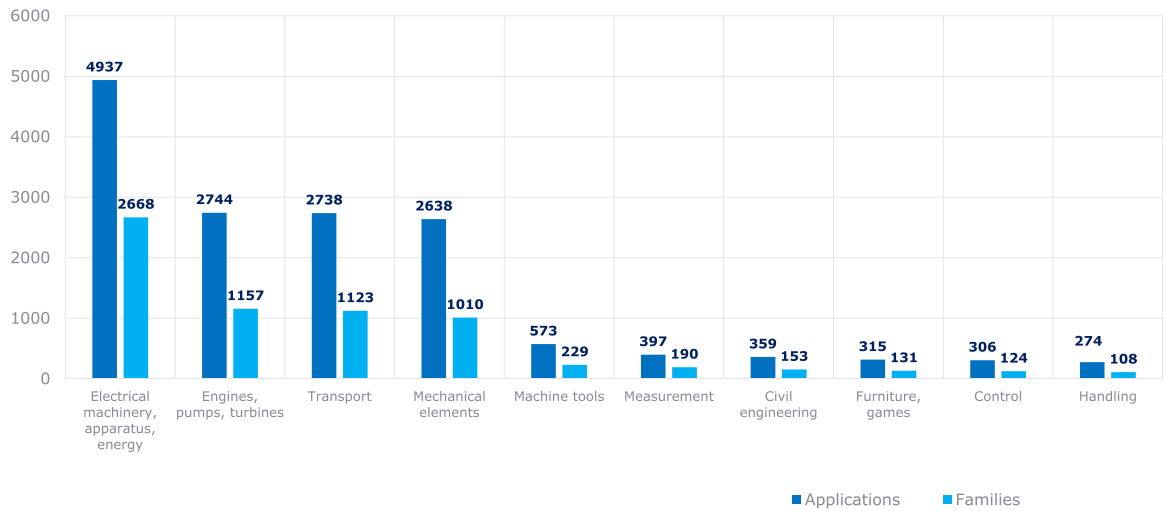


Fig. 10. Top 10 application fields with the highest number of patent families and applications in the flywheel area.

priorities, respectively.

The use of the international offices WIPO and EPO accounts for 17 %, approximately an eighth of the total number of publications, which reaffirms the local character of the area under study, with Germany, China, USA and Europe being the main markets of interest, as seen in the world map. This outcome highlights the countries with the highest activity in the most intense green color, revealing how Africa, Oceania and Latin America demonstrate almost no activity in the area of interest (Fig. 13).

From a market perspective, the pattern is the same: Germany maintains the leadership in the area with 20.7 % of total publications (46 documents are in effect), followed by China with 16.5 % of total publications (64 documents in effect). If the EPO and the countries

belonging to Europe are jointly considered, then they add up to 47.3 % of total publications. The Asian continent accounts for 25.3 % of the total number of publications. Specifically, Spain has 0 priorities and 13 publications, all of which were granted and 7 of which are in effect (Fig. 14). Finally, the figure summarizes the balance of publications in effect (green color) or that have entered the public domain (red color) in the first 20 offices. Specifically, China and the USA account for 43.5 % of the total number of documents in effect, and in Europe, taking into account the EPO and the countries belonging to Europe, a total of 126 documents are in effect. In addition, 289 documents (47.3 %) are still in the application or granted phase and therefore in effect, and the remaining 52.7 % have already entered the public domain.

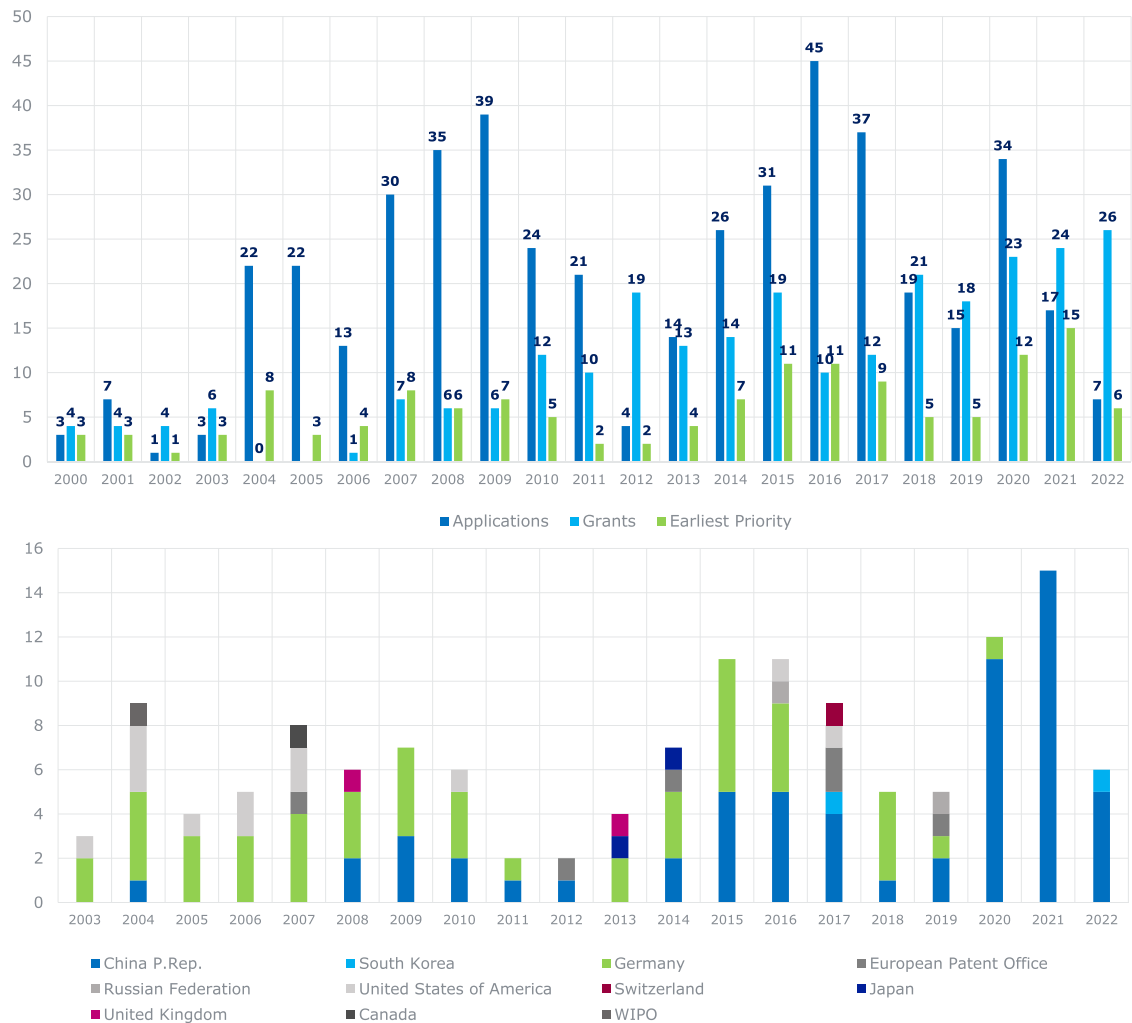


Fig. 11. Evolution of applications of shock absorbers patents, applications and grants and details of the 15 national and international offices with the highest number of priorities.

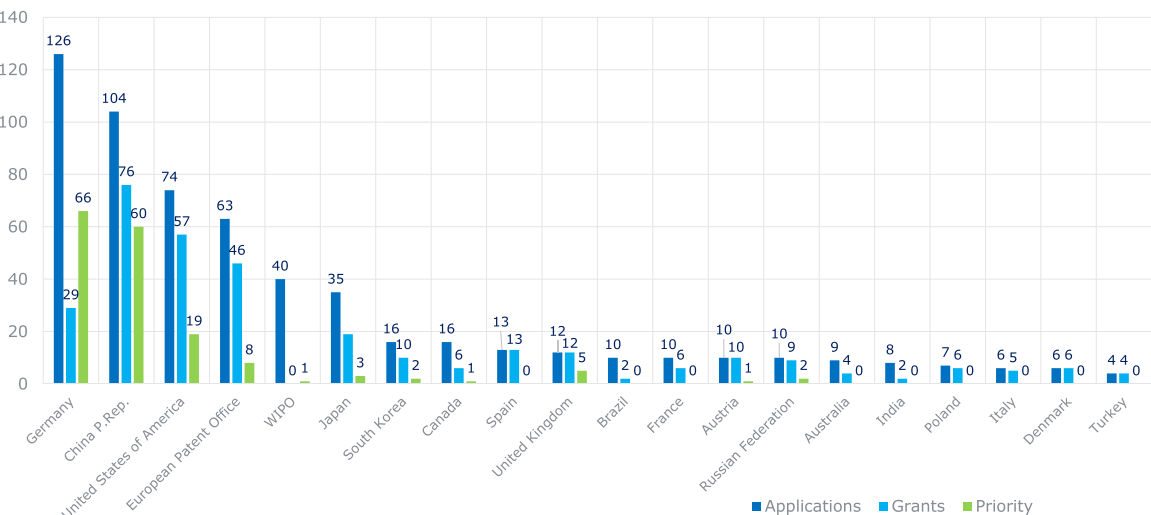


Fig. 12. Top 20 national and international offices ranked by publications, grants and patent applications.

3.2.3. Technological leadership

As in the previous section, two rankings of applicants are shown below (Fig. 15 and Fig. 16) the first refers to entities with the largest

number of families, and the second ranking refers to entities with the largest number of publications. In both sets, entities in different positions are highlighted, and their position in the rankings vary. In

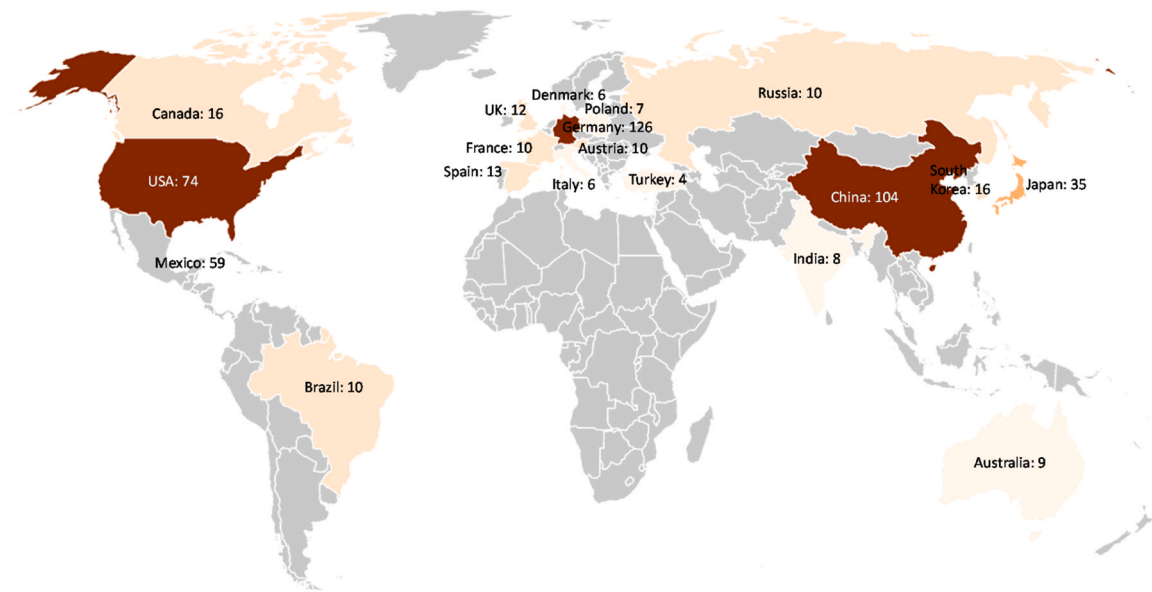


Fig. 13. World map of major national offices based on number of applications.

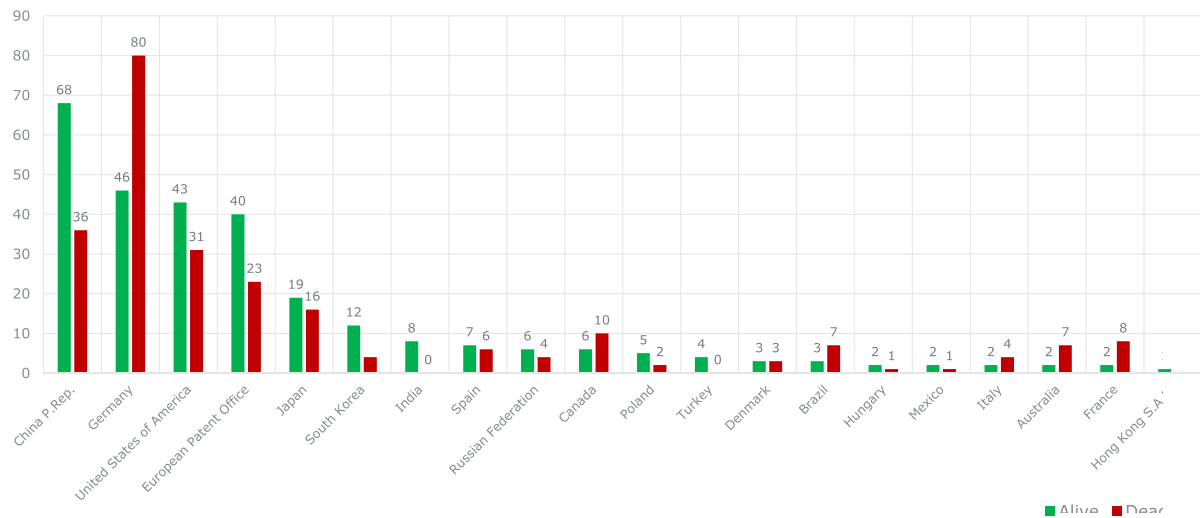


Fig. 14. Balance of the 20 national and international offices by validity of the patent documents and distribution of the total legal status of the documents collected.

addition, new players will also stand out, some of which have a high magnitude portfolio of innovations in the area, but in most cases such new players are not very widespread when compared to entities with few families and a very broad marketing strategy (e.g., multinational companies).

Simultaneously, it is important to point out that given the nature and participation of the applicants, the technology is fully introduced in the market, since 72 % of the applicants are companies, and the rest correspond to the participation of individual inventors and ultimately universities and research centers. With respect to the top 20 applicants with the highest number of patent families, European companies such as SCHAEFFLER AG (Germany), HYDAC (Germany), VOLKSWAGEN AG (Germany), Ypsomed AG (Switzerland), LUK LAMELLEN, and ROBERT BOSCH stand out in the first positions. However, the activity of research institutes, universities and individual inventors from the Asian continent is also evident. In the second ranking by the magnitude of publications, the German companies SCHAEFFLER AG and HYDAC remain in the top positions, but new players appear with few but widespread innovations such as GEAR L A INC and ADIDAS INT Marketing, RT FILTERTECHNIK GmbH, which register a family of widely extended patents.

Finally, the main fields of application of energy storage systems by shock absorbers in terms of innovations are shown in Fig. 17, according to the classification of the private Patbase database. These fields include mechanical elements, transport, pumps, turbines, electrical machinery and general technology.

4. Conclusions

This study analyzed the patent and innovation landscape of two mechanical energy storage systems, flywheels and RSA, over the period from 2002 to 2022 by collecting data on patents and utility models. The analysis of flywheels revealed that less than half of the total number of patents and utility models have been granted. China emerged as the leading innovator in this sector, followed by the United States, Japan, and Germany, which together account for 84 % of all patent families. The study also showed a continuous increasing trend in annual innovation applications and grants, indicating that while the technology is mature, it is still in a developing phase. Geographically, Asia-Oceania has generated almost two-thirds of all innovations, while Europe and America represent 15 % and 14 % of total priorities, respectively. From

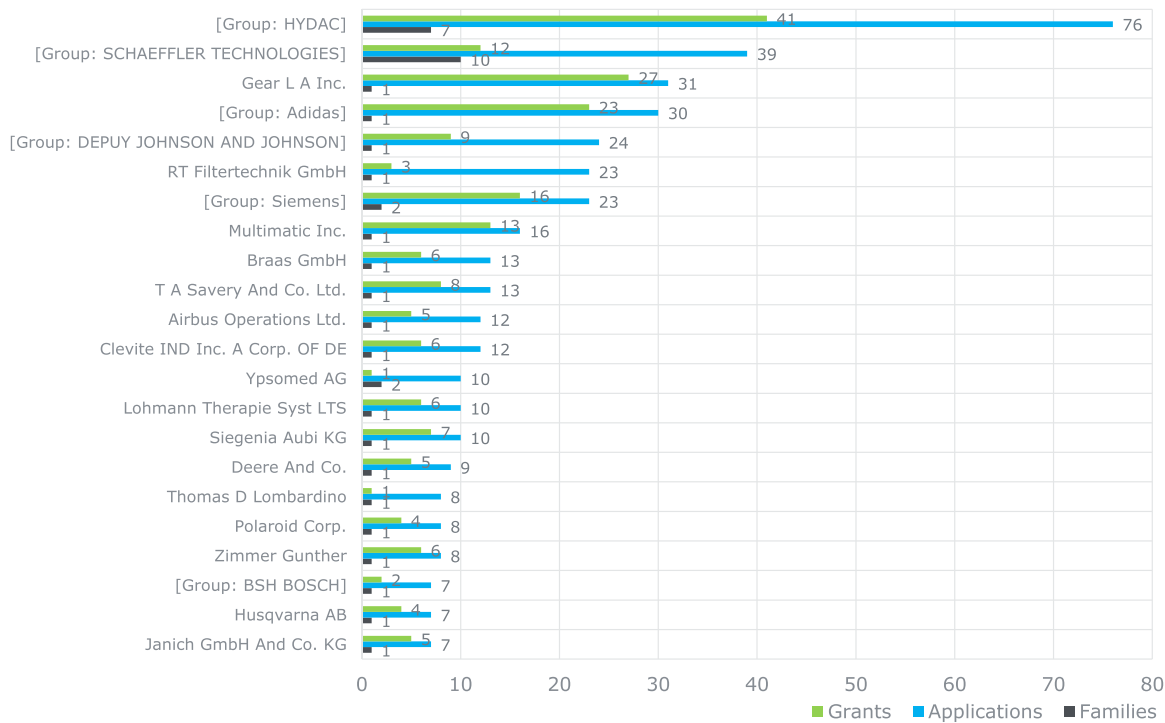


Fig. 15. Top 20 entities with the highest number of patent families.

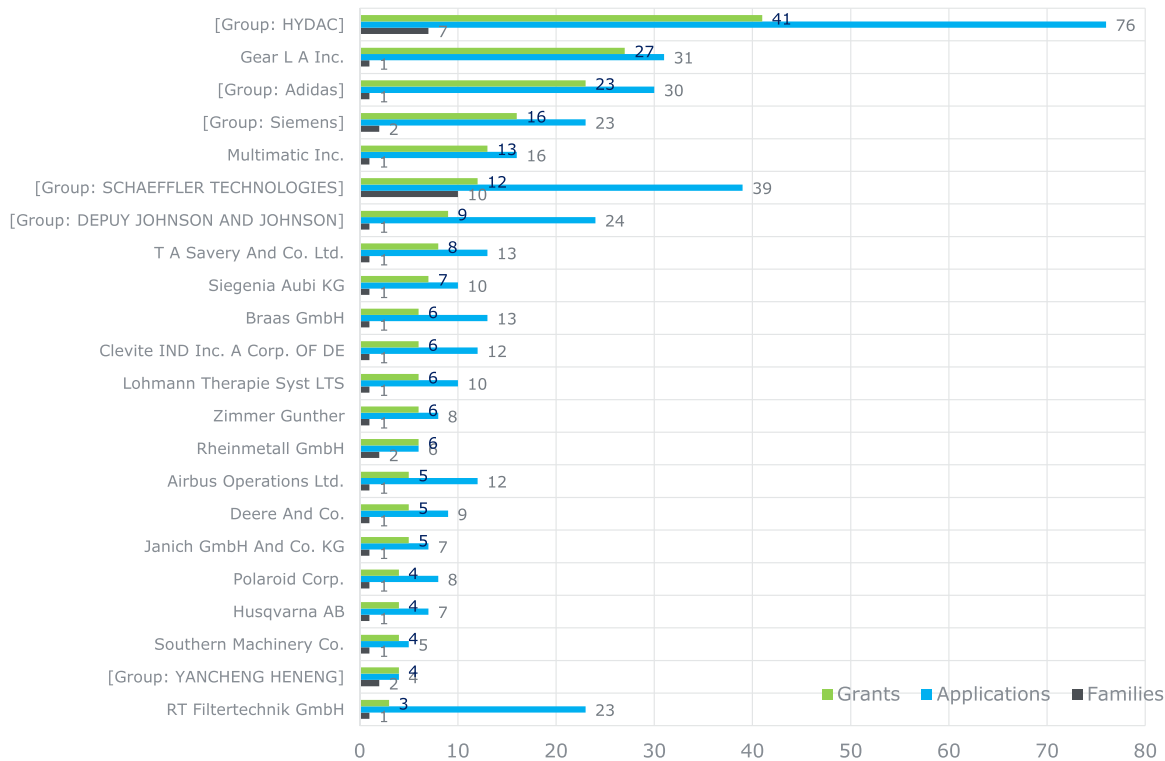


Fig. 16. Top 20 entities with the highest number of applications.

a market perspective, China maintains its leadership with 29 % of total publications, followed by the United States, Japan, and Germany. However, innovation activity is highly localized, focusing primarily on the Asia-Oceania, Europe, and North America regions. Leading companies and institutes in this field include Chinese firms such as CANDELA (SHENZHEN) TECHNOLOGY INNOVATION CO. AND

SHENYANG WEIKONG NEW ENERGY TECH CO., followed by German and American companies like LUK LAMELLEN, VYCON, SIEMENS, and CATERPILLAR INC. In the case of RSA, the study identified a total of 169 patent and utility model families and 599 publications, with 55 % of them granted. Germany, China, the USA, and the European Patent Office (EPO) were

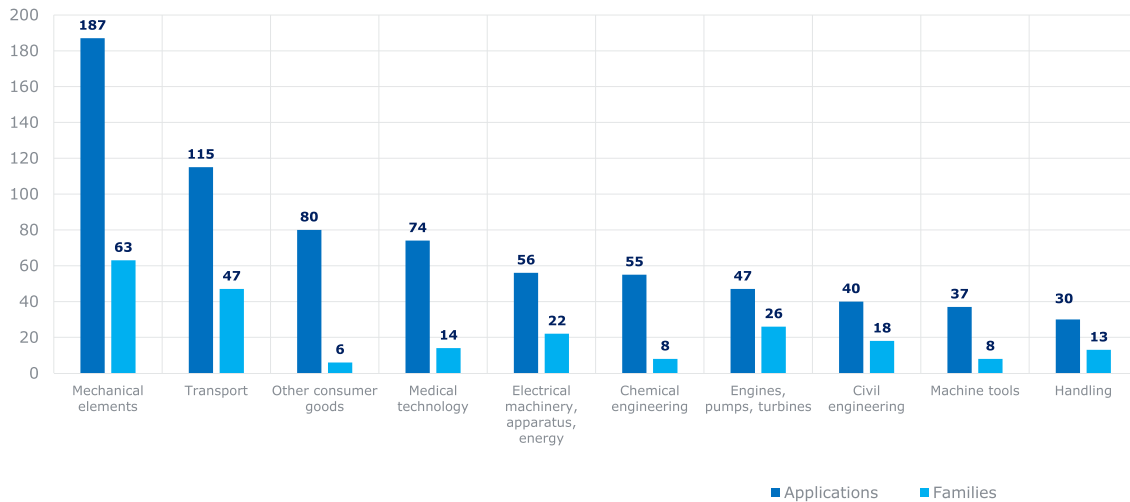


Fig. 17. Top 10 application fields with the highest number of patent families and applications in the area of regenerative shock absorbers.

found to be the leading countries in generating solutions in this area. The trend in annual innovation applications varied over the last two decades, peaking at 15 applications in 2021. Germany leads in knowledge generation, accounting for 39 % of the priority patents. From a market perspective, Germany, China, and Europe are the leading markets of interest, while Africa, Oceania, and Latin America showed almost no activity in this area. The study also identified the top 20 applicants with the highest number of patent families, which included European companies such as SCHAEFFLER AG, HYDAC, VOLKSWAGEN AG, and CONTINENTAL AUTOMOTIVE GmbH.

Future research should focus on enhancing cross-regional collaboration in the development of flywheel and RSA technologies, particularly between leading regions like Asia-Oceania and Europe. Integrating these mechanical energy storage systems with emerging technologies such as IoT, advanced machine learning, and AI could improve their efficiency and resilience. Addressing market expansion barriers in underrepresented regions like Africa, Oceania, and Latin America can broaden market reach and global impact. By addressing these areas, future research can advance, contributing to a more sustainable and efficient energy storage landscape.

CRediT authorship contribution statement

José Luis Míguez: Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Conceptualization. **Miguel Ángel Gómez:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Formal analysis, Data curation. **Sandra Rodríguez:** Writing – original draft, Software, Resources, Methodology, Investigation, Formal analysis, Data curation. **Jacobo Porteiro:** Writing – review & editing, Supervision, Methodology, Investigation, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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